Muhammad Ibrahim

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Education

New York University May 2026
Bachelor of Science in General Engineering (Mechanical and Civil); Major GPA: 3.60 (UK 2:1) *Abu Dhabi, UAE*
- Coursework: Solid Mechanics, Engineering Dynamics, Fluid Mechanics, Thermodynamics, Heat Transport

New York University Jan 2024
Study Away, Immigration and Theatre *Paris, France*

Experience

Center for Stability, Instability and Turbulence (SITE) May 2026 – Jul 2026
Graduate Intern *New York, USA*
- Delivered a free-surface model's complete training dataset of 60 trials at 4 angles of attack in 6 weeks, by running the water-channel rig solo.
- Tripled the model's valid range from 15 to 45 degrees, by grounding it in a dimensional analysis of surface height against flow velocity, submergence depth and body radius.
- Saved an estimated 50+ further tunnel runs, by training a Python and MATLAB model to predict free-surface profiles at untested angles.

Advanced Manufacturing Workshop Sep 2025 – May 2026
Workshop Technician *Abu Dhabi, UAE*
- Cut print failures from 50% to 10% over 70+ print hours, by diagnosing root causes, refining print parameters and checking critical dimensions with callipers.
- Raised part strength 50% over plain filament, by laying up bidirectional carbon fibre and fibreglass reinforcement.
- Delivered 8 load-bearing parts to 3 capstone teams with 15 unsound prints rejected before assembly, by running FEA and post-print dimensional checks.

Center for Stability, Instability and Turbulence (SITE) Apr 2023 – May 2026
Undergraduate Research Assistant *Abu Dhabi, UAE*
- Co-designed 5 water-tunnel model variants in SolidWorks for machine-shop manufacture, confirming boundary-layer separation in every run with flow visualisation.
- Built the centre's 130+ case fluid-instability dataset (Rayleigh-Taylor, Richtmyer-Meshkov) in a 2-person team.
- Prepared the dataset for a laminar and turbulent linearity study, by jointly post-processing the cases.

Projects

Advanced Anti-Roll Tank System (ART3S) — *SolidWorks, Simulink* Oct 2024 – Sep 2025
- Cut simulated vessel roll **85%** passively and by **more than 98%** with active control, by designing 200 to 800 m^3 anti-roll tanks for ferry and car-carrier hulls in SolidWorks.
- Shortened roll decay time **33%** passively and **62%** under active control, by modelling vessel and tank dynamics in Simulink as the design lead on an 8-person team.
- Presented the system to the Mubadala committee after testing it on a 29 cm scale hull at the Abu Dhabi Maritime Academy and the NYU Abu Dhabi wave tank.

Traffic-Powered Savonius Wind Turbine — *ANSYS Fluent, Composites* Sep 2024 – May 2025
- Raised peak moment **72%** and power **56%** over a non-serrated baseline, by adding serrated leading edges and a 45-degree helical twist to the blades in ANSYS Fluent CFD.
- Proved the concept with a 1:3 scale carbon-fibre-reinforced rotor that self-started within **2 to 5 seconds** in roadside testing, producing up to 10 V as vehicles passed.
- Projected 3 to 15 W at full scale for a turbine in a road's central reservation, harvesting the vortices from traffic passing on either side.

Extracurricular Activities

NYU Abu Dhabi Athletics, *Varsity Jiu-Jitsu, Recreational Sport and Team Videographer* 2022 – 2026
Habitat for Humanity International, *Engineer Volunteer, Dhulikhel, Nepal* 2025
Emirates Motorsports Organization, *Chief Track Marshal* 2019 – 2025

Skills

Design and CAD: SolidWorks (CSWA certified), mechanical design, system integration, rapid prototyping
Simulation: ANSYS Fluent (CFD), finite element analysis (FEA), Simulink, control systems, dimensional analysis
Testing and Manufacturing: water tunnel and wave tank testing, flow visualisation, FDM 3D printing, composites
Programming and Data: Python, MATLAB, machine learning, experimental data processing, root cause analysis
Languages: English (native), Urdu (native), Arabic (elementary)